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Isolating the effects of the cognitive interview techniques

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The cognitive interview (CI) is a procedure designed for use in police interviews involving witnesses. This study tested the most recent version of the CI (Fisher & Geiselman, 1992) which comprises not only 'cognitive' techniques such as context reinstatement but also 'social' techniques for increasing rapport. Children (aged eight–nine years) viewed a magic show and were interviewed after a short delay of two days (time 1) and/or a longer delay of 12 days (time 2). At time 1 the CI produced a significantly greater amount of correct recall than did a structured interview (SI) which was similar to the CI save for the CI's special cognitive techniques. However, at time 1 the CI also produced significantly more recall errors. At time 2, no differences occurred between CI and SI recall. There was a significant hypermnesia effect at time 2 for those groups interviewed twice suggesting an effect of retrieval practice. The time 1 effects of the CI were found to exist only in the questioning phase of the interview and social and cognitive explanations for the changes in the nature of recall with a CI are considered. Practical implications are discussed in the context of good practice for interviewing child witnesses.

he ability to obtain full and accurate information is critical in an investigation, yet ccording to the eyewitness literature, accurate and complete recall is difficult to chieve (e.g. Goodman, Aman & Hirschman, 1987). The cognitive interview (CI) as developed by Fisher and Geiselman in response to the many requests received om police officers for a method of improving recall in witnesses. It draws upon xperimental research on memory and is presented as a package of techniques that

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can be used to facilitate memory search and retrieval (Fisher & Geiselman, 1992). The present study attempts to identify the most effective components of the CI and compare it with current practices for interviewing children (see Bull, 1995*a*, *b*). The extent to which variables such as interviewer training, motivation and effective communication/rapport with the witness may contribute to gains in information are also explored. The research therefore addresses both theoretical and practical questions.

The CI procedure

The CI procedure essentially comprises four techniques and some strategies for improving communication in an interview (Fisher & Geiselman, 1992; Memon & Koehnken, 1992). One of the principal CI techniques is the mental reinstatement of the physical and personal contexts that existed at the time (see Memon & Bruce, 1985, for a review). Context reinstatement involves (*a*) emotional elements ('How were you feeling at the time?'), which may work via state-dependent effects (Eich, 1980), (*b*) perceptual features ('Put yourself back at the scene of the crime and picture the room; how did it smell, what could you hear?'), and (*c*) sequencing elements ('What were you doing at the time?'). The rationale for context reinstatement comes from the encoding specificity principle (e.g. Tulving & Thomson, 1973). The other CI strategies include instructions to search for details extensively (which can lead to the recall of additional relevant information; Geiselman & Fisher, 1988), to recount events in a variety of orders (Loftus & Fathi, 1985) and from a variety of perspectives (e.g. the perspective of the victim, suspect, another witness). These techniques are based on the assumption that memory trace inaccessibility is a result of a limited search.

The four techniques described above formed the basis of the original version of the CI. Further refinements of the CI (Fisher & Geiselman, 1992) include 'cognitive techniques' for activating and probing a witness' mental image of the various parts of an event, such as a suspect's face, clothing, objects etc. A distinction is drawn between conceptual image codes (an image stored as a concept or dictionary definition) and pictorial codes (the mental representation of an image; Paivio, 1971). In addition to the 'cognitive' components, the CI in its current form places considerable emphasis on social communication techniques, similar to those recommended to interviewers not employing the CI 'cognitive techniques' (Bull, 1992*a*; Memorandum of good practice, 1992). These techniques include 'transfer of control' of the interview from the interviewer to the witness. The latter is put into place during the rapport building phase in several ways. For instance it allows witnesses to dictate the pace of the interview and structure their own recall. This is achieved through use of open questions, by not interrupting witnesses, by timing questions carefully so that they are related to witnesses' retrieval patterns and not to a protocol that an interviewer may be using. For example, if a witness is describing a suspect's face, an appropriate question would be to ask about eye colour rather than to ask about the suspect's shoes. These social communication strategies are said by Fisher & Geiselman (1992) to facilitate the effective implementation of the cognitive CI techniques described earlier. For example, by not interrupting witnesses while

ey are attempting to recreate context, and by pausing between questions, the terviewer allows the witness time to form an image and to engage in a more haustive search, and this may induce more elaborate responses.

In early tests of the CI, the interviewers (students or police officers) received some struction in use of the 'cognitive' components of the CI and collected memory ports of simulated events that students had witnessed two days earlier. In the rst study, instructions were given to interviewees in a written form so inter-ewer–witness interaction was minimal. With relatively brief training, interviewers otained up to 35 per cent more correct information without an increase in errors, compared to a no-training control group (e.g. Geiselman, Fisher, MacKinnon & olland, 1985, 1986). When the revised version of the CI was tested it was found to enerate even more information than the original CI (Fisher, Geiselman, Raymond, urkevich & Warhaftig, 1987). These results were taken as strong evidence of the fectiveness of the CI. However, upon closer examination of the studies it was pparent that there was one major drawback, namely, the lack of a suitable control roup.

ontrol group

rom a practical perspective, it is important to show that the CI is more effective than ie techniques currently recommended for use by police officers and others who nduct investigative interviews. However, an effective control group is needed to emonstrate that the CI cognitive techniques themselves are causing the positive ffects and not other factors such as social communication, quality of questioning, apport building skills or interviewer (or interviewee) motivation. The use of the rm 'standard' in earlier studies itself implies inferiority. In the present research we ought to construct a control interview (the structured interview; Koehnken, Thürer Zorberbier, 1994) in which the quality of training in communication and uestioning techniques was comparable to the CI and which followed that ecommended to professionals who interview children (e.g. Memorandum of good ractice, 1992). The essence of the Memorandum's guidelines on interview structure to treat the interview as a procedure in which a variety of interviewing techniques re deployed in relatively discrete phases proceeding from free recall to open and then pecific, closed-form questions. Rapport building, active listening and not nterrupting (transfer of control) are also emphasized as important components. The I techniques fit nicely into this basic framework.

he CI and children

iven current concerns about the vulnerability of children in criminal proceedings .g. Goodman & Bottoms, 1993) and the skills of those who interview child itnesses (Bull, 1992; Clyde, 1992), it is timely to focus on the utility of the CI with is group. In earlier tests, six–seven-year-old children did not show any advantage hen interviewed with a CI as compared with a structured interview (Memon, ronin, Eaves & Bull, 1993; Memon, Cronin, Eaves & Bull, 1996). A subsequent

study suggested that young children had difficulty using some of the CI strategies (e.g. the 'reverse order' recall instruction). This could be due to developmental limitations (Flavell & Wellman, 1977) or task demands (Cronin, Memon, Eaves, Küpper & Bull, 1992). More recently, the CI procedure has been further modified for use with child witnesses. Saywitz, Geiselman & Bornstein (1992) evaluated the CI using seven–eight-year-olds and 10–11-year-olds as witnesses to a live event. A 'practice session' was included (Expt 2) to familiarize the children with the interview techniques. The interviewers were college students (practice interview) or experienced police officers (interview about main event) who received written instructions and a two-hour training session which included information about child-appropriate language, rapport building, interview preparation and procedure, and information on the use of the four original cognitive components of the CI. In the control (standard) condition police officers were instructed to use the techniques they would normally use. The CI led to the recall of more correct details as compared with the standard interview without increasing errors. The CI benefited the older children more than the younger children.

The CI has been investigated in Germany in several studies designed to examine separately any effects of the CI on (*a*) the incidence of errors and (*b*) confabulation (Koehnken, Finger, Nitschke, Höfer & Aschermann, 1992; Koehnken, Schimmossek, Aschermann & Höfer, in press; Koehnken *et al.*, 1994; Mantwill, Aschermann & Koehnken, 1995). With adults, there was an increase in correct details and errors with the CI in two studies (Koehnken *et al.*, in press; Mantwill *et al.*, 1995). With children (aged 9–10 years), it was found the CI did substantially increase the amount of correct information recalled. However, confabulations (the report of details not present in the event) also increased significantly (Koehnken *et al.*, 1992). Several more recent studies where the children have ranged in age from seven–nine years have also reported increases not only in correct information but also in errors with the CI (McCauley & Fisher, 1995; Milne, Bull, Koehnken & Memon, 1995). From a theoretical and practical perspective it is clearly important to identify why the errors occur and whether they can be minimized.

Repeated testing with the CI

So far, research has primarily examined the effects of CI under optimal encoding conditions (a single interview after a relatively brief delay). In the real world a witness may be interviewed many days after the event. Studies of children's delayed recall show that inaccuracies increase over time (Flin, Boon, Knox & Bull, 1992; Poole & White, 1993). Also, a witness may be interviewed more than once. While there has been concern addressed about the possible effects of repeated poor interviews with children (Clyde, 1992), laboratory studies of memory and memory development have documented two main positive effects of repeated testing using a variety of stimulus materials. The first is 'reminiscence', or the recall of material that did not appear in an earlier test. When this new information exceeds the amount of information that is forgotten, the 'hypermnesia' effect has occurred (Payne, 1987). Both these phenomena have been demonstrated in eyewitness contexts using videotaped (Scrivner & Safer, 1988) and staged criminal scenarios (Turtle & Yuille,

994). We know that memory improves for a short period of time after exposure Brainerd, Reyna, Howe & Kingma, 1990). Another advantage of repeated testing nortly after an event is that it inoculates against forgetting (Brainerd & Ornstein, 991; Warren & Lane, 1995).

It is predicted that an initial CI interview will produce hypermnesia effects at econd interview on the basis that its techniques such as context reinstatement ncrease overlap between encoding and retrieval conditions and enrich connections etween episodic traces and semantic features (Brainerd *et al.*, 1990). These factors ill improve children's performance on a second test relative to those who had no rst test or an inferior first test. An alternative explanation of increased reporting ith CI following repeated testing would be that the CI merely shifts a witness' riterion for how much to report. (Note that one of the CI instructions to witnesses to report everything, even that which they are unsure of). Multiple retrieval ttempts with the CI may act as conversational training (Hudson, 1990). Poole White (1995) made the useful point that a forensic interview is both a memory est and a conversation, and so it is difficult to predict which of these two aspects will nost influence recall.

The conclusions of previous CI studies are contradictory. Some have found an ncrease in correct information, some find an increase in errors and others get no ffects (see Koehnken, Milne, Memon & Bull, 1994, for a review). Moreover, it may e possible for an interviewer armed with a range of 'good' interviewing techniques nd effective communication skills to achieve the same results as a CI interviewer. he present study aimed to determine, through careful matching of experimental and ontrol group, whether any CI effects reflect improved retrieval searches or whether ney are a result of improved interviewer–witness communication. The primary aim f the present study is to test this hypothesis and to assess the nature of inaccuracies nat may occur when the CI technique is used with children. In order to directly ompare the results of this study with previous research, children aged eight–nine-ears are used. Previous research has shown this age group show a smaller advantage f testing with CI as compared to older children (Saywitz *et al.*, 1992) and make more rrors in their reports (Koehnken *et al.*, 1992; McCauley & Fisher, 1995; Milne *et al.*, 995).

Method

articipants

One hundred and nine children aged between eight and nine years of age participated in the study. There ere 53 males and 56 females.[1] There were 29 cognitive interviews and 32 structured interviews erformed at time 1. The number of children in each of the four repeat testing groups was as follows: I/CI = 10; CI/SI = 16; SI/CI = 15 and SI/SI = 13. In addition, the following numbers of children ere interviewed at time 2 only: NCI (none, CI) = 16 and NSI (none, SI) = 17.

[1] Of the 109 children, nine children were absent for testing; in addition, two interviews (one SI, one CI) were iscarded without coding on the basis that the children had made no attempt to talk about the event. One further nterview (CI) was dropped from the sample because it appeared as an outlier with an extremely unusual number of onfabulations ($N = 78$). Several CI and SI interviews from time 1 and time 2 were not transcribed as the tape ecordings were of poor sound quality. This reduced the total number of interviews used in the analysis to 87.

The event

Children took part in a magic show, an event chosen because of its interest to children. A local magician was contacted and agreed to take part in the study. The magic show lasted for approximately nine minutes and there were 12 performances to small groups of children. The magic show was videorecorded.

The magic show was performed to groups of between eight and 12 children (mean = 9) over a period of two mornings in a room (not their classroom) which the children, although familiar with, did not use often. After a delay of between one or two days (equally distributed across the conditions) some of the children were interviewed in school using a cognitive or structured interview. The remainder were not interviewed at that time. After a further delay of between 10 and 13 days (equally distributed between the conditions) all of the children were interviewed (some now for a second time) by either a cognitive or structured interviewer. The children interviewed twice had different interviewers at each time.

Design

A 3 (cognitive, structured or no first interview) by 2 (cognitive and structured) design was used to examine the effects of interview technique and repeated testing.

Interviewer training

Four interviewers were trained solely in the cognitive interview and four in the structured interview. The cognitive interviewers were two male and two female students. The structured interviewers were three female and one male student. The cognitive interviewers and structured interviewers were trained separately by a highly qualified researcher (GK) with many years of interview training experience. The structure and content of the training for each group will be described fully as this is something earlier researchers have failed to do. Similarities and differences between the SI and CI should be noted.

Each group of interviewers was trained in two four-hour sessions that began with an introductory lecture on the importance of the interview in psychological assessment and information gathering in various situations. Both groups were given some guidelines about non-verbal behaviour in the interview (e.g. seating position, eye contact, pauses and speech rate). For each step in the training there was a demonstration role play excerpt (a CI/SI child witness interview conducted by a trained interviewer) which was followed by several live practice role plays (interviewers were asked to choose an event, playing the part of a child and interviewer) and these were video recorded. Individual feedback was given. There were plenary discussion and question sessions at the end. In addition to active role plays, the interviewers were encouraged to rehearse mentally the various stages of the interview. For each group the interview was divided into the following phases.

Rapport. Boggs and Eyberg (1990) pointed out that the essential first phase of the interview is to establish rapport between child and interviewer. She is asked to describe a familiar event, for example, a favourite game. Follow-up comments such as 'that sounds fun, tell me how you play it?' increase rapport and prime the child to give elaborate responses. Both the CI and SI group practised rapport building in this way (cf. Saywitz *et al.*, 1992). An important part of the rapport building was the transfer of control from interviewer to interviewee (which included active listening, not interrupting and effective use of pauses). As part of this transfer of control the interviewer makes it clear that he or she does not have the information about the event but rather it is the child who holds the information. This procedure typically forms part of the CI but it has also been advocated in other interviewing guidelines (e.g. the Memorandum of good practice, 1992) and therefore this component formed part of the training for the SI as well as the CI group.

Free recall phase. The SI interviewers were asked to request a free narrative account from the witness and this was used as a strategy for obtaining information in the subsequent questioning phase. The CI group

received identical instructions for the free recall phase, but in addition they were given training in encouraging witnesses to reinstate the context mentally (as described in the introduction) before they began. The CI interviewers also employed the 'report everything' instruction at this stage.

Prompt phase. At the end of the free recall phase, CI and SI interviewers paused briefly and used one prompt, 'Please tell me more', before commencing the questioning phase.

Questioning phase. In the next phase both the CI and SI interviewers were asked to use the information reported by the witness in their free recall phase as a guide for follow-up questions. Both the CI and SI interviewers were instructed in the use of appropriate types of questions. They were asked to begin with open questions and then follow these with closed questions. In general interviewers were asked to use the free recall to find out who was present at the event and what they did. Where a person was mentioned, interviewers were asked to elicit details about clothing. They were specifically instructed to avoid leading, misleading and forced-choice questions. The CI interviewers received additional training in the activation and probing of images relating to various parts of the event. (For example, the children were told to 'picture the magician's face, and then describe it').

Second retrieval phase. The purpose of this phase was to examine the effects of additional instructions on the recall of new information. The second retrieval phase of the interview was different for the two groups. The CI interviewers employed the CI 'reverse order' recall instruction at this point. This took the following form: 'Tell me about the very last thing you remember in the magic show and then what happened before that, and before that, so you're working your way back to the first thing you remember.' It was placed towards the end of the interview so that any extra information it elicited could be identified. The SI group also attempted to elicit additional information at this stage by asking children to go through the event again, recalling additional details if possible.

Both groups received a summary of the theoretical background material relevant to their training and a detailed handout containing all the training material to study. Each group was led to believe they were the 'experimental group'.

Coding and scoring of the interview transcripts

Using the videos of the 12 magic shows, two research assistants identified the details that could be recalled. This produced some 650 details. The information contained in the verbatim transcript of each interview was checked with the corresponding video and classified as correct, as an error (that is, wrongly describing something that was present or did happen) or as a confabulation (e.g. saying a 'pig' was present when it was not).

Information was classified into four detail types describing persons, actions, objects or surroundings. Although any classification of detail type is dependent on the sort of event used, there were several concerns that we wanted to address. Firstly, we wanted to test the hypothesis that the CI, by encouraging reinstatement of context, merely increases the reporting of details about surroundings (Memon *et al.*, 1996). The present coding classification therefore includes a separate count of such details. Secondly, we wanted to examine whether the CI might increase the reporting of all types of erroneous information, or whether errors (if they occurred) may reflect the difficulty children have in giving descriptions about persons as compared to actions (Davies, Tarrant & Flin, 1989).

The stage at which information was reported in the interview was noted with a distinction being made between free recall (FR), prompt (PR), questioning (QU), and second retrieval (SR) phases. After the information contained in the child's free report was scored for accuracy, recall appearing in the other subsequent sections of the interview (QU, PR and SR phases) was only scored if it was new. This is illustrated in Appendix 1.

Two research assistants coded and scored the interview transcripts and any uncertainties were resolved by discussion. In addition to this, 10 transcripts were coded by both of the research assistants and inter-coder reliabilities were calculated: inter-coder agreement for total accurate was 96 per cent ($r = .93$, $p < .0001$) and for total errors 89 per cent ($r = .86$, $p < .001$).

Results

Two types of measures are reported throughout. Firstly, the absolute number of errors and confabulations are reported. Secondly, instead of simply reporting the absolute number of correct details, these are expressed as a proportion of that available to be recalled because each of the 12 magic shows contained slightly different information (i.e. descriptions of the children who helped, what they were wearing etc.). This will be referred to as 'percentage correct' in the results described below.

As is now the convention in research on the CI, the number of correct, incorrect, and confabulated details are also each expressed as a proportion of the total number recalled. The proportion is calculated by dividing the number of each type of detail (i.e. correct, or incorrect and confabulations) by the total number of details reported (i.e. correct plus incorrect plus confabulations). This measure is usually referred to as accuracy (and sometimes as reliability).

First interviews

Amount recalled at time 1. A series of one-way ANOVAs was performed to test the prediction that the CI would increase the amount of correct and of incorrect information (errors and confabulations)[2]. For percentage correct recall there was a significant effect ($F(1, 59) = 6.17, p < .05$) with the CI producing more correct recall. Similarly, the absolute numbers of errors was significantly higher in the CI condition ($F(1, 59) = 4.17$, $p < .05$). There was no significant effect of type of interview on confabulated details ($F < 1$). The increase in percentage correct information with the CI was a difference of 20 units of information. This was accompanied by an increase of 3.31 units of erroneous details. So with a control group matched in every respect to the CI save the 'cognitive' components of CI, there is a positive effect on the percentage of correct recall. There is also an increase in the absolute number of errors (although not in the proportion of recall that was inaccurate) and this is considered at length in the discussion.

Accuracy at time 1. It was found that, on average, 88 per cent of the children's recall was correct with 8 per cent errors and 4 per cent confabulated details. This is consistent with earlier research (Saywitz *et al.*, 1992). A one-way ANOVA (interview type) was performed on the accuracy of recall (amount correct expressed as a function of total amount reported). From this no significant differences in accuracy emerged across interview type ($F < 1$).

Recall at each phase of the interview at time 1. The 'percentage correct', total number of correct details and absolute amount of inaccurate information reported at each phase of the interview can be seen in Table 1. Most of the accurate information the children

[2] MANOVAS were not performed on the data as the dependent variables did not always correlate (see Cole, Maxwell, Arvey & Salas, 1994).

recalled was in the FR and QU phases of the interview, as would be expected given information was only scored later if it was new.

Table 1. Percentage correct recall, total correct and errors by interview phase (time 1)

	Cognitive interview			Structured interview		
	% correct	Total correct	Errors	% correct	Total correct	Errors
Free recall						
M	7.5	48.1	2.1	7.3	46.4	2.2
SD	4.4	28.2	2.3	3.8	24.0	2.0
Prompt						
M	0.7	4.5	0.1	0.2	1.2	0.0
SD	1.8	11.38	0.6	0.6	3.5	0.2
Questioning						
M	6.6**	42.7	6.8**	4.0	25.7	4.0
SD	2.8	18.0	4.7	2.3	14.8	3.2
Second retrieval						
M	1.0	6.5	0.9	1.3	8.1	0.9
SD	1.3	8.4	1.5	1.3	8.5	1.3
Total						
M	15.8*	101.9	10.0*	12.7	81.5	7.1
SD	4.9	31.9	6.6	4.9	31.4	4.6

* $p < .05$; ** $p < .01$.
Note. The 'percentage correct' measure shows the percentage of *possible* accurate recall for event.

Previous research with child witnesses has not established where in the interview the effects of CI appear. In the present study the CI instructions were given prior to both the FR and QU phases. As indicated earlier, information from the FR phase was used to probe for further detail in the QU phase. On this basis it would be expected that the CI would show an increase in new information at each of these phases. A series of one-way ANOVAs were performed to look at the 'percentage correct' information, errors and confabulations at each phase of the interview.[3]

The only phase to yield significant effects of interview type was the question phase. The percentage correct information was significantly higher in the CI as compared to the SI group ($F(1, 59) = 16.22$, $p < .001$). This was accompanied by a significant increase in errors ($F(1, 59) = 8.43$, $p < .01$). There was no effect of interview type on confabulations in the QU phase ($F(1, 59 = 1.32$, $p > .05$; means = 3.62 for the CI and 2.59 for the SI).

[3] Combining all the phases in one repeated measures ANOVA was deemed inappropriate as each phase involved different recall techniques, and recall at each phase was scored on the basis of what was recalled at the earlier phases (additional information). The same principle was adhered to regarding detail types.

One question that was of particular interest in this study was the effects of multiple retrieval attempts within a single interview session on recall. We were interested in whether the reverse order recall (CI) instruction could generate any more information than a simple request to try again. There was no beneficial effect of using the CI reverse order recall instruction over a simple instruction to go through the event one more time. In other words, there were no significant CI–SI differences in the SR phase ($F < 1$). Averaging across these two conditions, there were approximately six additional units of information generated within the additional retrieval attempt.

Recall by type of detail at time 1. So far the analyses show that the significant CI–SI difference in percentage correct and errors only occurred in the questioning phase. It was important to examine the question phase in more depth to ascertain the types of details (person, action, object and surrounding) being recalled. The next series of analyses breaks down the 'percentage correct' recall and errors at the QU phase into types of information recalled (see Table 2) and a series of one-way ANOVAs was conducted to compare performance across CI and SI for each dependent variable (confabulations are not reported in this section as there were no significant main effects).

Table 2. Percentage correct information, total correct and errors in the QU phase by detail type (time 1)

	Cognitive interview			Structured interview		
	% correct	Total correct	Errors	% correct	Total correct	Interview: Number of errors
Person						
M	7.5	8.7	3.4*	5.7	6.6	1.9
SD	4.7	5.5	3.0	3.7	4.4	1.5
Action						
M	6.2**	22.7	1.7	3.1	11.5	1.1
SD	4.0	14.9	1.8	2.8	10.2	1.9
Object						
M	8.6**	10.8	1.8	5.3	6.7	1.0
SD	4.3	5.4	1.9	3.8	4.8	1.4
Surrounding						
M	1.6	0.55	0.0	2.8	1.0	0.0
SD	2.8	0.95	0.0	4.5	1.5	0.0

* $p < .05$; ** $p < .01$.
Note. The 'percentage correct' measure shows the percentage of *possible* accurate recall for event.

The person category was of most interest in the present study as it is known that both adults (MacLeod, Frowley & Shepherd, 1994) and children (see Davies *et al.*, 1989) have difficulty in describing people accurately (Clifford & Bull, 1978). An

ANOVA revealed a significant effect of interview type on person errors in the QU phase ($F(1, 59) = 6.38$, $p = .01$) with the CI having significantly more person errors in this phase. Contrary to prediction, there were no effects of interview type for person correct ($F(1, 59) = 2.65$, $p > .05$).

Taking action details next, the CI was found to increase percentage correct information ($F(1, 59) = 11.77$, $p < .001$) but not action errors ($F(1, 59) = 1.89$).

Finally, for objects there was an effect of interview type on percentage correct in the QU phase ($F(1, 59) = 5.40$, $p < .01$) with more correct information in the CI. There was no significant effect for object errors ($F(1, 59) = 3.37$, $p = .07$) or details about surroundings ($F < 1$) across conditions.

Number of interviews and event knowledge. The eight interviewers (four cognitive and four structured) conducted between five and 10 interviews at time 1 and nine and 12 interviews each at time 2. As the interviewers were unfamiliar with the event when they started interviewing at time 1, it was important to check that they were not eliciting more information at the end of their series of interviews than at the beginning, due to their increased knowledge of the event. For each interviewer, interview position (first interview, second interview etc.) was correlated with amount of accurate information obtained. There were no significant correlations (interviewer A, Fisher's $r(5) = -.09$; interviewer B, $r(8) = -.16$; interviewer C, $r(6) = .24$; interviewer D, $r(9) = .16$; interviewer E, $r(10) = .10$; interviewer F, $r(8) = .34$; interviewer G, $r(6) = .24$; interviewer H, $r(7) = -.09$).

To summarize the results of the first interview, those children having a CI interview at time 1 recalled more correct information than did those having an SI. They also made more errors. However, accuracy rates did not differ. The significant effects occurred in the QU phase and examination of performance in this phase revealed that in the CI group there was more correct recall about actions and objects and more person errors.

Second interviews

When analysing the data yielded at time 2 a series of 2 × 3 ANOVAs were performed to examine CI–SI differences at time 2 taking into account type of interview at time 1 (CI/SI/none).

Amount recalled at time 2. Examination of percentage correct, amount of errors and amount of confabulations revealed a significant effect of interview type on percentage correct information only ($F(2, 81) = 5.38$, $p < .01$). *Post hoc* analyses revealed that those children who had either a CI (Fisher's PLSD = 3.10, $p < .05$) or an SI (Fisher's PLSD = 4.43, $p < .01$) at time 1 recalled significantly higher percentage correct at time 2 than those children who were not interviewed at time 1 (see Table 3 for means). There were no significant differences in accuracy of recall. The mean accuracy rate (across all conditions) was 84 percent.

Having identified an increase in percentage correct information in the groups who experienced two interviews, analyses were performed to look at differences at each stage of their second interview (as with the time 1 data).

Table 3. Percentage correct information, errors and confabulations means at time 2

	Time 1					
	CI		SI		None	
Time 2	% correct	Total correct	% correct	Total correct	% correct	Total correct
CI						
M	16.4	105.4	18.1	115.7	13.5	86.5
SD	4.0	26.5	6.9	44.5	6.7	42.8
SI						
M	15.6	100.5	16.3	104.5	12.2	78.2
SD	5.5	35.3	4.3	28.2	3.2	20.1

Errors	CI	SI	None
CI			
M	16.2	12.2	11.2
SD	7.9	7.6	6.9
SI			
M	15.0	16.3	11.3
SD	8.5	12.4	11.0

Confabulations	CI	SI	None
CI			
M	6.8	7.4	4.1
SD	10.3	6.8	4.6
SI			
M	5.7	8.4	9.3
SD	7.1	19.4	13.0

Note. The 'percentage correct' measure shows the percentage of *possible* accurate recall for event. Due to dropouts at time 2, the *N* for CI and SI at time 1 were 26 and 28.

Recall at each phase of the interview (*time* 2). When a series of 3×2 ANOVAs was performed on the data, with percentage correct, errors and confabulations at each phase of the interview as the dependent measures, the following results were obtained. Taking the FR phase first, ANOVA revealed a significant effect of first interview on percentage correct free recall ($F(2, 81) = 7.06$, $p < .01$). *Post hoc* analyses showed that the children who had no interview at time 1 recalled a lower percentage of correct information during the free report phase of their interview at time 2 than those who had either a CI (Fisher's PLSD = 3.73, $p < .01$) or an SI (Fisher's PLSD = .25, $p < .01$) at time 1. No other significant effects were noted in the FR phase.

In the QU phase, there was an effect of first interview on percentage correct ($F(2, 81) = 4.11$, $p < .05$). Contrary to prediction, those children having an SI interview at time 1 reported more correct information in the QU phase at time 2

compared to those having a CI interview at time 1 (Fisher's PLSD = 1.82, $p = .01$) and compared to those having no interview at time 1 (Fisher's PLSD = 1.44, $p < .05$). There were no other significant effects in the QU phase and no significant effects in the SR phase.

Proportion of new and repeated information at time 2. The coding scheme developed for the second interviews enabled the information to be categorized as either new (not mentioned at time 1) or repeated information (mentioned at time 1 and repeated at time 2). For those children having two interviews ($N = 54$), the amount of repeated and new information and the proportion of their recall that was either new or repeated were noted. The amount of new and repeated information did not vary as a function of interview method. Interestingly at time 2, 59 per cent of accurate information was repeated and 41 per cent was new; 23 per cent of errors were repeated and 77 per cent were new; 12 per cent of confabulations were repeated and 88 per cent were new. There are many possible reasons for the new information at time 2 (including the children having talked amongst themselves and to others between interviews). It is worthy of note that a number of script-relevant intrusions occurred at the second interview and this may reflect the influence of schema on gaps in memory (see Milne *et al.*, 1995; Wark, Memon, Koehnken & Bull, 1995, for specific examples).

Number of questions asked and recall performance. Both groups of interviewers (CI and SI) were instructed to ask questions concerning only what the children had told them during the free report phase of the interview. As stated above, no differences in the amount recalled during the free report phase were found between those children having a CI and those having an SI at time 1 or time 2. However, in this phase the CI interviewers asked significantly more questions at time 1 than did the SI interviewers ($t(58) = 8.06$, $p < .0001$, means for CI = 30.18 and for the SI = 12.88). Of these questions 66 per cent were open questions, 22 per cent closed and the remaining 2 per cent fell into the leading or multiple choice category. The same pattern was found at time 2 but here there were no CI/SI differences. The greater number of questions asked by CI interviewers suggests that, when faced with the same amount of free recall information as their SI counterparts, the CI interviewers subsequently adopted a more detailed questioning strategy.

Discussion

At time 1 (two-day delay) the present study found the CI to produce an increase in the amount of correct recall. The effect size ($d = .58$) was somewhat smaller than that reported in earlier studies (see Koehnken *et al.*, 1994) but the control group (the structured interview) is superior to that used in the majority of previous studies. The increase in correct recall with the CI emanated from significantly more accurate object and action details. The increases in correct recall and errors were manifested in the questioning phase of the interview and this raises a number of questions about the effects of techniques used in this stage of the interview.

To what extent is the increase in reporting of details in the questioning phase of

the CI due to the use of specific CI techniques? The CI interviewers used a combination of context reinstatement and imagery instructions in the questioning phase. The interviewers actively encouraged the children to generate images of the event and to describe them. They prompted the children with specific questions based on the information the children had given in free recall. The two techniques, activation of images and detailed questioning, could account for the gains in correct and incorrect details. While imagery may be used to facilitate retrieval of information from memory, it may also increase inaccurate reporting as can be seen in the literature on suggestibility and creation of false memories (Ceci, Loftus, Leichtman & Bruck, 1994; Hyman & Pentland, 1996). Similarly, the use of specific questions with children of the age group sampled in the current study may increase the amount of information that is reported but at the cost of a drop in accuracy (Davies *et al.*, 1989; Dent & Stephenson, 1979; King & Yuille, 1987; Oates & Shrimpton, 1991).

If we look at the absolute numbers of errors compared to correct details, it is clear that the CI produced a greater increase in correct details than in incorrect details. One plausible explanation for this is that the CI technique (which includes 'report everything') shifts the criterion for responding by influencing confidence and willingness to report information (the theoretical basis is illustrated by signal detection theory). Bekerian and Dennett (1993) drew attention to this in a review paper which assessed the theoretical basis of CI. The effect of CI on the response criterion requires further investigation. It is possible that the errors are a result of demand characteristics or social pressure to give a desired response (Cronin *et al.*, 1992). Asking probing questions can have this effect (Davies *et al.*, 1989; Dent, 1991; Oates & Shrimpton, 1991). This study was, however, able to locate the nature of the errors that were being made. The children made more errors about persons than any other type of information. Again previous literature suggests that children have difficulty in describing persons (Davies *et al.*, 1989; Gee & Pipe, 1995).

How should the increase in errors be interpreted by practitioners? Firstly, it should be noted that the CI increases errors but not confabulated details. An increase in confabulated details (e.g. saying the magician wore a cloak when there was no cloak) may have quite different ramifications in a forensic context than a person error (e.g. describing a purple jacket as a blue one). Secondly, if we look at the absolute number of correct details compared to errors in the present study, for every six correct details, there was an error, so the gains appear to outweigh the risks. Thirdly, there is no difference in accuracy rates across cognitive and structured interview conditions. While procedures such as the CI aim to increase the amount of information that is reported, in a forensic context it is the accuracy of the information that is crucial (Koriat & Goldsmith, 1994). In a recent review, Koriat & Goldsmith (1996) demonstrate how critical it is to separate the measure of quantity of information which fits the 'storehouse metaphor' of traditional research from applied research on everyday memory processes. We suggest that the CI be used with older children (Saywitz *et al.*, 1992) but some caution be exercised when interviewing younger children. Whenever possible external corroboration should be sought for the details reported. Special care should be taken in eliciting person descriptions as several studies have found errors with this type of information in a CI (see also Milne & Bull, 1995; Milne *et al.*, 1995). Finally, it may be the case that the effects obtained

here are specific to the eight–nine-year-old age group. We would encourage researchers to test the effectiveness of the CI procedure described in this study with older and younger children.

While the time 1 results were consistent with our hypotheses, the time 2 results were not. In light of the evidence that the CI (at time 1 and time 2) occasions a more extensive memory search, we predicted an increase in additional and repeated information in the CI/CI group at time 2. Similarly, we predicted a carryover effect boosting performance of the CI/SI group. This was not found; as there was no effect of CI at time 2 (although performance was better if children had been interviewed at time 1). A number of recent studies have obtained similar results (Brock & Fisher, 1994; McCauley & Fisher, 1995). There are several possibilities as to why this may be the case. Taking the position that CI achieves its effect by enhancing interviewer communication and interviewee expectations, perhaps a good first (SI) interview is sufficient to raise performance levels (Roediger & Payne, 1982). It is possible that, over the delay, memory for the event became less context dependent (cf. Smith, 1988). This would not be incompatible with recent theoretical accounts of memory and forgetting (e.g. Brainerd *et al.*, 1990; Riccio, Rabinowitz & Axelrod, 1994; Roediger & Challis, 1989). Similarly, Payne, Hembrooke & Anastasi (1993) maintain that context is a poorer cue at the delayed test than at immediate test either as a consequence of a decrement in context–item association strength, or as a result of changes in the functional context between the two sessions.

The improvements following repeated testing suggest a powerful effect of retrieval practice. If a retrieval attempt in first interview serves to strengthen item-to-context associations sufficiently, such associations would tend to be recovered on the second test (Brainerd *et al.*, 1990). As indicated in the introduction, it is well established in the memory literature that testing shortly after exposure to the to-be-remembered material will attenuate forgetting over a delay (see Brainerd & Ornstein, 1991, for a review). The extent to which quality of the first interview may determine the amount reported upon repeated testing may be explored in future studies by including an untrained control group.

Further research could also be usefully deployed to look at conditions under which repeated interviews are most likely to be effective. From a practical and theoretical perspective, it is important to understand how performance with a CI will vary with longer delays, to examine the patterns of losses and gains (e.g. the frequency with which repeated and new information is reported) and factors that may reduce inter-test forgetting. Interviewer credibility, demand characteristics, interviewer instructions and an interviewee's interpretation of the interviewer's requests may also interact in interesting ways with interview technique and these may account for changes in response criteria across repeated tests. Memory theorists have recently developed experimental procedures (e.g. the 'logic of opposition' procedure; see Jacoby, Woloshyn & Kelley, 1989; Lindsay, Gonzales & Eso, 1995) which can be applied to address these issues.

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Appendix 1

Sample scored transcript (cognitive condition)

The following is an extract taken from one interview transcript to illustrate the coding and scoring procedure that was used in this study. The data is presented in a table form for illustration purposes only. In actuality, we worked with an entire interview and checked off details against a master list comprising of all details reported by all participants. Only *new* information was scored at each stage, i.e. if an item was mentioned in free recall it was not scored again in the question phase (e.g. 'bald').

Free recall. The man looked baldish and I know exactly what happened about the five pound note. He got it behind him the note and just pushed it up.

He done the bird trick and he had a thingy over it and a cover down all round it and um he took it out and there was a bird there, and that's how he done it.

There's me and Matthew and there's the magic man....

Prompt. He had two handkerchiefs and he tied them together and I blowed them and they come undone again. And a pigeon one and a rabbit one....

Questioning. He wore some like a uniform trousers, a tie and he had a bald head. He was pretty tall and was wearing brown shoes and all the children were sitting by the doors. And after we done the tricks, we went and sit down for a while and then Russel came up....

Second retrieval attempt. He said, 'Can I have two helpers and that boy over there and that girl over there' and I went up. He says 'Can you hold the string on each side' and his scissors were stiff but the magician cut the rope....

Information	Correct[a]	Error	Confabulation
Free recall			
Man	1		
Baldish	1		
Five pound note	2		
Bird trick	1	1	
Cover	1		
Something appeared	1		
Matthew (not there)			1
Prompt			
Hankies	2		
tied	1		
Blows on hankies	2		
undone	1		
Rabbit	1		
Question			
Trousers	1		
Tie	1		
Brown shoes	1	1	
Russel	1		
Second retrieval			
Asks for helpers	1		
Two children	2		
Hold string	2		
Scissors	1		
Magician cuts rope	3		

[a] A point was given for every detail based upon how much information was provided. For example, 'five pound note' received two points because it provided more information than the response 'money'.